PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

GROUP : YOLOO!!

Members:

1) Mohammad Afiq Zakwan Bin Asri
2) Akmal Hariz Bin Azidi
3) Adib Rusyaidi Mohd Zaki
4) Nur Khadijah Aleya Binti Azizul
5) Nur Dianah Binti Amir Chong

Project Name: SmarTax - AI Tax Decision Copilot

Version: 1.0

1. Project Overview

- **Problem Statement:** Every year, more than 8 million Malaysian individuals and 1.2 million SMEs file income tax returns, and most of them overpay. LHDN publishes over 200 pages of Public Rulings covering reliefs, rebates, and exemptions, but these documents are written in legal language, scattered across the hasil.gov.my website, and never surface inside the e-Filing portal itself. As a result, working individuals routinely miss the Lifestyle relief, SSPN relief, PRS relief, parental medical relief, and education-insurance relief they are legally entitled to claim. Small and medium enterprises pay RM 2,000–5,000 annually to accountants largely because they cannot verify which expenses are deductible under Section 33 of the Income Tax Act 1967 or which capital allowances apply under Public Ruling 12/2014. Freelancers, a rapidly growing segment of the Malaysian workforce, fall through the cracks entirely because existing tools assume either pure employment income or full corporate structure. The core issue is not calculation as LHDN already provides an e-Filing calculator. The real issue is decision intelligence: taxpayers do not know what they qualify for, what trade-offs exist between competing reliefs (PRS vs SSPN vs additional EPF), or what actions they can still take before 31 December to legally reduce their tax liability for the current year of assessment.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence — specifically, personal and small-business financial intelligence with a focus on taxation: relief eligibility discovery, tax estimation, scenario-based trade-off analysis, and year-end tax planning recommendations grounded in Malaysian law.

- **Proposed Solution Summary:** SmarTax **i**s an AI-powered tax decision copilot that reads a user's EA form (for employees), profit and loss statement (for SMEs), or mixed-income records (for freelancers) and returns (1) a complete tax estimate with a full breakdown, (2) a personalised list of reliefs the user qualifies for but has not

claimed, (3) a pre-year-end action plan ranking concrete steps by tax-saving per ringgit of liquidity required, and (4) a natural-language chat interface for asking tax questions. Every recommendation is grounded in a specific, citable LHDN Public Ruling via retrieval-augmented generation (RAG), and the deterministic tax calculation itself is performed by an auditable TypeScript engine, never by the language model. This architectural separation means users get the interpretive flexibility of an LLM with the mathematical trustworthiness of a calculator. If Z.AI's GLM is removed from the system, SmarTax can no longer extract structured data from unstructured EA forms or P&L statements, cannot reason across user profiles and LHDN rulings to surface eligible reliefs, cannot generate personalised action plans, and cannot answer natural-language tax questions in Bahasa Malaysia or English with cited sources. What remains would be a generic tax calculator , which already exists and has not solved the underlying problem.

2. Background & Business Objective

- **Background of the Problem:** For decades, personal and SME tax filing in Malaysia has followed a pattern shaped by three structural constraints. First, the rules themselves live in long-form legal documents , the Income Tax Act 1967, Schedule 6, and LHDN Public Rulings — which ordinary taxpayers never read. Second, the LHDN e-Filing portal is primarily a submission portal: it collects data the taxpayer already knows, but it does not proactively advise the taxpayer on what they could claim. Third, the professional advice layer (tax agents and accountants) is priced outside the reach of most individuals and many small SMEs. The outcome is a large, persistent information asymmetry: ordinary taxpayers systematically overpay, while those with professional advisors systematically underpay.

  Recent advances in large language models, particularly those capable of multilingual reasoning over legal-domain text, have made it feasible for the first time to bridge this asymmetry at near-zero marginal cost. The LHDN rulings exist as public documents; the tax bands are public; the user's own data is already in their EA form or accounting records. The missing piece has always been personalised reasoning, and that is now solvable.

- **Importance of Solving This Issue:** Malaysians collectively leave hundreds of millions of ringgit on the table each year by failing to claim reliefs they are entitled to. For an individual earning RM 78,000 per year, claiming the full Lifestyle, SSPN, PRS, and Medical Insurance reliefs translates directly into approximately RM 3,000 of tax saved , money that stays in their household. For an SME owner, systematically claiming Section 33 business expenses and the correct capital allowances can reduce annual tax by tens of thousands of ringgit. On a national scale, closing this information gap has meaningful welfare implications: it improves disposable income for B40/M40 households and improves cash-flow resilience for small businesses, which are both policy priorities under the Madani economic framework. Solving this issue also professionalises the experience of taxation for a generation that expects financial tools to be instant, conversational, and mobile-first.
- **Strategic Fit / Impact:** SmarTax aligns directly with the Domain 2 problem statement of UMHackathon 2026: enabling individuals, businesses, and communities to make timely and informed decisions by replacing fragmented data and intuition-based processes with AI-driven insight. It also aligns with Z.AI's strategic positioning of the GLM model as a reasoning engine capable of structured output, multilingual fluency, and grounded retrieval, all of which are load-bearing for tax decisions. At the product level, SmarTax is designed as a horizontal platform: the same architecture that serves an individual filer today can serve an SME tomorrow and an accounting firm after that, creating a clear path from hackathon MVP to real-world commercial viability.

3. Product Purpose

3.1. Main Goal of the System

To give every Malaysian taxpayer, such as individual, freelancer, or SME owner a trustworthy AI tax decision partner that explains, in plain language, exactly what they owe, what they qualify for, and what they can still do to pay less, with every recommendation traceable to a specific LHDN Public Ruling.

3.2. Intended Users (Target Audience)

Primary users (MVP focus):

- Salaried individual taxpayers (B40/M40 income brackets)
- Small and medium enterprise (SME) owners filing corporate tax
- Freelancers and gig-economy workers with mixed income

Secondary users (future expansion):

- Licensed tax agents and accounting firms
- HR teams distributing tax-filing tools to employees
- University students and young professionals filing for the first time

Primary personas driving the MVP design

- **Aminah, 28 — Marketing Executive.** Salary RM 6,500/month. First-time filer two years into her career. Owed RM 1,800 last April and had no idea whether the figure was correct. Doesn't know SSPN, PRS, or lifestyle relief exist. Represents the first-jobber segment.
- **Encik Razak, 45 — Owner of Razak Enterprise Sdn Bhd.** Revenue RM 1.8M/year. Pays his accountant RM 3,500/year but suspects deductions are being missed. Needs to understand capital allowances, entertainment expense limits, and schedule 3 treatment of small assets.

- **Jia Wen, 32 — Freelance Designer.** Mixed employment with freelance income. Unclear whether to register as sole proprietor, how to deduct her laptop and internet, and whether she needs to file Form B or Form BE. Represents the growing gig-economy segment.

4. System Functionalities

4.1. Description

SmarTax operates as a guided, three-stage decision-support pipeline. Stage one (ingest) accepts unstructured input which is a PDF or photo of an EA form, a P&L statement, or a conversational interview answered through a short wizard, and uses GLM to produce a clean, structured user profile. Stage two (reason) takes that profile, retrieves the most relevant LHDN rulings from a vector database seeded with 15 Public Rulings, and asks GLM to identify every relief the user qualifies for along with a natural-language justification and an exact citation. Stage three (decide) passes the reasoning output to a deterministic TypeScript tax engine that applies 2026 tax bands, relief caps, and rebate logic, then asks GLM to generate a ranked pre-year-end action plan. Throughout, users can also ask tax questions in a chat, with answers based on the same LHDN rulings.

4.2. Key Functionalities

- *Smart Relief Finder*

A conversational wizard asks eight lifestyle questions (marital status, number and ages of children, parents supported, SSPN account holders, lifestyle purchases, insurance premiums, zakat contributions, EV charging subscriptions). GLM maps these answers against the retrieved LHDN rulings and returns the exact list of reliefs the user qualifies for, each with a plain-language reason and a clickable citation.

- *Deterministic Tax Estimator with Full Breakdown*

A pure TypeScript engine applies 2026 tax bands, computes chargeable income, applies relief caps, subtracts rebates (zakat and the RM 400 individual rebate where chargeable income is below RM 35,000), and subtracts any PCB already paid. The result screen shows the complete waterfall: gross income → deductions → reliefs → chargeable income → tax bands → rebates → final tax owed or refund due.

- *Pre-Year-End Action Plan*

For each user, GLM generates three ranked recommendations for actions they can take before 31 December to reduce their tax for the current year of assessment. Each action is ranked by tax-saving-per-ringgit-of-liquidity-required and includes an exact deadline and a ruling citation. This is the decision-intelligence heart of the product: it converts knowledge into action.

- *Ask SmarTax Chat*

A natural-language chat interface where users can ask open-ended questions such as "Can I claim my MacBook as a freelancer?" or "Is my solar-panel installation tax-deductible?" Every answer is RAG-grounded in the ruling corpus and ends with a citable source.

- *Bilingual Interface (Bahasa Malaysia and English)*

Full interface translation and GLM response localisation, respecting the user's preferred language. Recommendations, citations, and action plans all render natively in Bahasa Malaysia when that mode is selected.

- *Mode Switcher (Individual, SME, Freelancer)*

A single codebase serves three modes with different relief databases, different tax-band logic (individual progressive versus SME 15%/17%/24% bands), and different GLM prompt templates. Users choose their mode on the landing page.

4.3. AI Model & Prompt Design

4.3.1 Model Selection

SmarTax uses Z.AI GLM (glm-4) as its only reasoning engine. We chose it for three reasons that matter to our product:

- **Handles two languages at once.** LHDN rulings mix English legal terms with Malay words, and Malaysian users often switch between English and Bahasa Malaysia in the same sentence. GLM understands both without needing a separate translation step.
- **Gives clean, structured answers.** Every step of SmarTax needs GLM to reply in a fixed format (for example, a list of reliefs or extracted tax fields). GLM follows format instructions reliably, so our code can read its answers without breaking.
- **Read long documents without losing focus.** To find the right reliefs, GLM has to read up to 5 ruling excerpts plus the user's profile at the same time. It handles this amount of text well and sticks to what's actually written in the rulings instead of making things up from memory.

4.3.2. Prompting Strategy

SmarTax uses a step-by-step prompting approach that gives GLM a clear role, real source material, and a fixed output format. We combine three techniques:

1. Give GLM a clear role with strict rules. Every request starts with a system prompt that tells GLM it is a Malaysian tax advisor. It must only suggest reliefs found in the rulings we provide, and it is not allowed to invent ruling codes. This keeps GLM behaving consistently across all four parts of our product.
2. Feed GLM the real rulings, not its own memory. For relief finding and chat, we pull the most relevant LHDN rulings from our database (using pgvector) and paste them into the prompt. This way, GLM reasons from the actual law instead of guessing from what it learned during training. This cuts down on made-up answers and makes every result easy to check.

3. Force GLM to reply in a fixed format. Every prompt that feeds into other parts of our code asks for a specific JSON structure. If GLM returns broken JSON, our code notices and either asks GLM to fix it, or falls back to a safe default answer.

This approach fits a decision-intelligence product because it keeps GLM's job small and testable, keeps answers based on real law, and keeps a clean line between what GLM does (understanding and explaining) and what our code does (calculating numbers).

4.3.3. Context & Input Handling

SmarTax enforces explicit input limits at every stage to keep inference costs predictable and outputs stable.

Input type	Limit	Action when exceeded
Retrieved rulings context	Top 5 chunks × 500 tokens = 2,500 tokens	If the search finds more than 5 matching ruling paragraphs, only the top 5 most relevant results are kept.
Ask SmarTax chat message	1,000 characters	Longer messages are truncated with a UI warning; the user is invited to split the question.
Chat conversation history	Last 6 turns kept in context	Turns beyond the window are dropped; a summary of older context is generated by GLM and re-injected if the conversation continues.

4.3.4 Fallback & Failure Behavior

SmarTax never blindly trusts GLM's answers. We assume any response could be wrong and have backup plans at every step. These are intentional design choices, not last-minute fixes:

- **Check the format, retry once if broken.** Every GLM response must follow a specific JSON structure. We check this using Zod. If the format is wrong, we ask GLM one more time: "Your previous answer was not valid JSON. Please try again using this exact format." If it fails again, the user sees a friendly error message and we log the problem.
- **Flag low-confidence extractions.** When GLM reads an EA form, it must also rate how confident it is ("high", "medium", or "low"). If confidence is "low", we pause the flow and ask the user to review and correct the numbers manually — instead of quietly using numbers that might be wrong.
- **Block made-up rulings before they reach the user.** Before showing any relief recommendation, our system checks whether the LHDN ruling code that GLM mentioned actually exists in our database. If GLM made up a ruling that doesn't exist, we hide that relief and record the mistake in our logs.
- **Keep the calculator working even if GLM goes down.** If the Z.AI API is offline or overloaded, users can still use the tax calculator by selecting their reliefs manually. The AI features show a "temporarily unavailable" message instead of crashing the whole app.
- **Politely refuse non-tax questions.** If a user asks something unrelated to tax (like "what's the weather?"), GLM is told to say it can only help with tax topics and points the user back to the features it supports.

5. User Stories & Use Cases

5.1. User Stories

The following user stories drive the MVP feature prioritisation. Each is written in the standard "as a / I want / so that" form so that acceptance criteria are unambiguous.

- **As Nina (first-time filer),** I want to input my EA form details and be told exactly which tax reliefs I qualify for, so that I don’t have to read hundreds of pages of LHDN rulings to understand what I can claim.
- **As Nina,** I want to see, in plain language and in Bahasa Malaysia, why each relief applies to me, so that I can explain my filing to my parents or verify with a friend.
- **As Nina,** I want to know what I can still do before 31 December to reduce my tax, so that I don’t miss year-end deadlines for SSPN deposits or PRS contributions.

- **As Encik Razak (SME owner),** I want to input my company P&L details and see which expenses are deductible versus non-deductible, so that I can review my accountant's work with confidence.
- **As Encik Razak,** I want the system to flag capital-allowance claims I may have missed under Public Ruling 12/2014, so that I stop leaving capex deductions on the table.
- **As Jia Wen (freelancer),** I want to ask open-ended tax questions such as "Can I claim my MacBook?" in natural language, so that I get an answer with a citation instead of guessing or paying a consultant.
- **As Jia Wen,** I want to slide through different Private Retirement Scheme(PRS) contribution amounts and see how each affects my tax, so that I can choose the contribution level that matches my cash flow.
- **As any user,** I want every recommendation linked to a specific LHDN Public Ruling I can click on, so that I can verify the advice independently and defend my filing if questioned.

5.2. Use Cases (Main Interactions)

Use Case 1 — EA Form Input Detail sand Tax Discovery

The user lands on the homepage, selects "Individual" mode, and input their EA form details. SmarTax parses the document with GLM, pre-fills an editable summary for confirmation, then presents the wizard. After the wizard is completed, SmarTax runs the relief-finder pipeline and the deterministic calculator, and displays the full tax breakdown alongside the ranked list of reliefs found.

Use Case 2 — Pre-Year-End Action Planning

From the results screen, the user clicks "What can I still do this year?" SmarTax generates three ranked actions with estimated savings, amount needed to spend, deadlines, and

LHDN citations. The user can mark an action as "I'll do this" and it gets saved to their filing record so they can check back later.

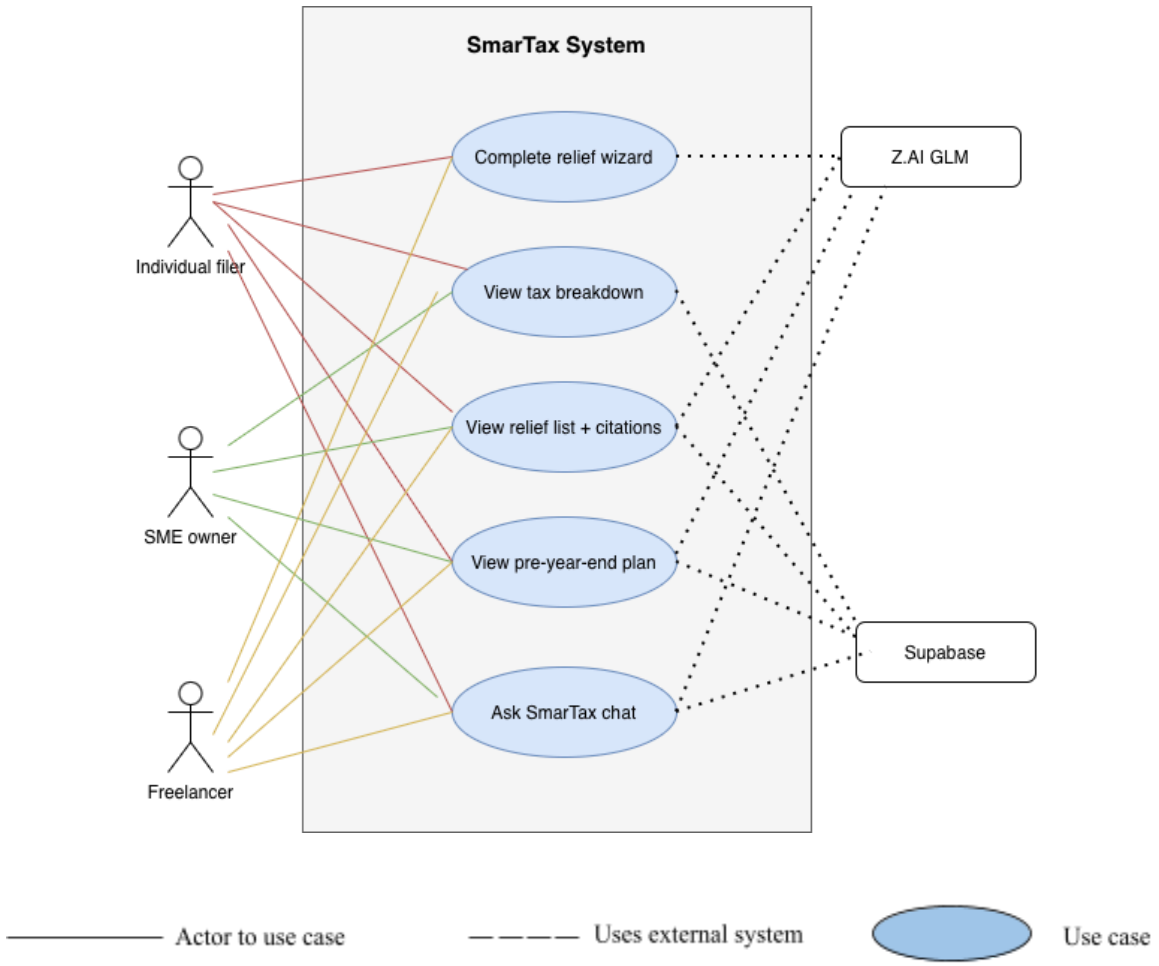
Use Case 3 — Ask SmarTax Chat

Jia Wen types "Can I claim my MacBook as a freelancer?" SmarTax retrieves Public Ruling 12/2014 and Section 33 of the ITA, passes both into GLM with her freelancer profile, and returns a cited, bilingual answer. Jia Wen can then click the citation to see the full ruling excerpt.

Use Case 4 — Mode Switch (Individual to SME)

Encik Razak switches from Individual mode to SME mode via the top-bar selector. The same core architecture reloads with the SME relief database, SME tax bands (15% first RM 150k, 17% next band, 24% above), and SME-specific prompt templates, without requiring a separate codebase.

5.3. Use Case Diagram



6. Features Included (Scope Definition)

- **An 8-question wizard that collects personal details** like marital status, children, insurance, and lifestyle purchases to find matching reliefs.
- **Smart Relief Finder** that shows which tax reliefs the user qualifies for, with plain-language explanations and real LHDN ruling citations.
- **A trustworthy tax calculator** built with pure code (no AI) that shows the full breakdown from gross income to final tax owed.
- **Pre-year-end action plan** that recommends 3 things the user can do before 31 December to legally pay less tax.
- **Mode switcher** that lets users choose between Individual, SME, or Freelancer, each with its own relief list, tax bands, and GLM prompts.
- **One-click language toggle** between English and Bahasa Malaysia for the entire app and all AI responses.
- A **chat feature where users can ask any tax question** in their own words and get answers backed by LHDN rulings.
- **Results dashboard with a visual waterfall chart** showing how the final tax figure was calculated step by step.

7. Features Not Included (Scope Control)

The following features are explicitly deferred to post-hackathon releases. They are listed here to make scope control visible and testable.

- **Direct submission to LHDN e-Filing.** SmarTax is positioned as a decision-support tool, not a filing service, to sidestep licensing and liability concerns.
- **Real-time bank and e-wallet integration.** Open banking connectors are a known next step but require PDPA compliance work beyond the hackathon window.

- **Year-round receipt wallet with photo capture and auto-categorisation.** Roadmapped as a post-MVP differentiator but not required for the preliminary-round demo.
- **Native iOS and Android applications.** The MVP is a responsive web app installable as a PWA; native apps will follow once PWA metrics justify the investment.
- **Accountant-firm multi-client dashboard.** The enterprise tier serving CPA firms is a planned revenue stream but out of scope for the preliminary round.
- **Automatic ingestion of LHDN Budget updates.** The ruling database is seeded once for the MVP; the continuous-update pipeline is a future enhancement.
- **Multi-user family or household mode.** Each account represents one filer in the MVP.
- **Mandarin and Tamil localisation.** BM and EN cover the largest user base for the MVP; additional languages follow after user validation.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Each user session uses roughly 18,000 to 24,000 GLM tokens, costing only a few cents per session. To keep costs low, we cache repeated chat questions so the same answer is served without calling GLM again. We also limit prompt sizes to a fixed length so every call uses a predictable amount of tokens.
- **Technical Constraints:** The MVP is a web app only , no native iOS or Android app. Tax bands and relief caps are hard-coded for Year of Assessment 2026. Only 15 LHDN Public Rulings are loaded in the database, so reliefs outside this set will not appear. Z.AI GLM is the only AI model used , if the API goes down, AI features show "temporarily unavailable" but the tax calculator still works.

- **User Input:** SmarTax cannot work without the user inserting their EA form or P&L details and answering the wizard questions. Users are responsible for checking the final numbers before filing. SmarTax is a decision-support tool, not a filing service, so it requires the user's own judgement before taking action.

9. Risks & Questions Throughout Development

- **Made-up reliefs or citations:** What if GLM invents a ruling code that doesn't exist? Our system checks every ruling code against our database before showing it to the user. If the code doesn't exist, that relief is hidden and the issue is logged.
- **Users thinking SmarTax is official tax advice:** What if users treat our recommendations as legal advice? The app clearly states it is a decision-support tool, not a filing service. Disclaimers remind users that final responsibility is theirs or their accountant's.
- **GLM API going down during demo:** What if we run out of API quota during the hackathon? We limit how many calls each user can make, cache common questions, and have a pre-recorded backup demo ready in case the API fails.
- **Privacy and data protection (PDPA):** What if someone else sees a user's tax data? Supabase row-level security makes sure users can only see their own filings. Uploaded files are stored securely and automatically deleted after 30 days.
- **How much ruling text to retrieve:** Should we pull small paragraphs or full sections from LHDN rulings? Currently set to small paragraphs (500 tokens each) for better accuracy, but still testing.